

InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

Aris Dressino^{a,*}

^a*TUM School of Management, Technical University of Munich, ArcisstraSSe
21, Munich, 80333, Germany*

Abstract

This paper presents *InventOR*, a prompt-configured large-language-model (LLM) workflow for material-level inventory decision support under demand and lead-time uncertainty. A material-scoped historical CSV and inline parameter block are supplied through an enterprise LLM application; no project-specific model weights were trained or fine-tuned on enterprise data. The LLM returns a policy artifact that deterministic parsing and simulation evaluate against an SAP-derived comparator and an SAP-safety-lead-time-informed operations-research (r, Q) comparator. The LLM is therefore a candidate-policy generator, not an autonomous ERP controller. The executed artifact set covers all 365 eligible plant-material pairs in a three-plant industrial dataset after retry handling and capacity validation. A common-feasible cohort of 346 pairs applies identical MOQ and per-material storage rules to all arms. Working-day post-cutoff simulation reports 77.52% demand-weighted aggregate fill, 97.94% mean material fill, and 2,848.96 mean simulated material-horizon holding/order cost for the LLM-emitted arm. Positive lost-sales penalty sensitivities remove its

*Corresponding author. Telephone: +49 89 289 25000.
Email address: aris.dressino@tum.de (Aris Dressino)

apparent cost advantage. These values remain descriptive simulated comparison rather than controlled production-trial evidence. Stored metadata do not establish the exact deployed prompt or model/runtime identity; the contribution is a transparent prototype workflow and disclosure of controls required for prospective evaluation.

Keywords: Generative Artificial Intelligence, Large Language Models, Operations Research, Supply Chain Management, Decision Support Systems, Explainable AI, Inventory Control

1. Introduction

Supply chain planning operates under volatile demand, uncertain replenishment, and disruption exposure [1]. Many material-control decisions nevertheless still depend on simple parameters: reorder points, safety stock, order quantities, review periods, and sometimes safety lead time; these settings strongly affect service and inventory exposure [2, 3]. Classical operations research offers rigorous tools for stochastic and robust inventory decisions [4, 5, 6], but operational planners often need a narrower capability: converting imperfect historical records into transparent, defensible, and auditable material-level policy values [7].

Large language models (LLMs) can help connect business intent, operational evidence, and analytical workflows [8, 9]. In inventory control, however, recommendations require explainability, reproducibility, accountability, and meaningful human control before planners can act on them [10, 11]. This creates a practical design requirement for hybrid AI-supported OR systems: LLM artifacts must be inspectable, while their numeric recommendations require explicit simulation and human release controls [9, 11].

18 This paper addresses that requirement through *InventOR*, an operations-
19 research-first workflow for material-level inventory decision support. An
20 enterprise LLM application produces policy artifacts from material-scoped
21 inputs; deterministic software parses and simulates their numeric values.
22 The study evaluates independent plant-material pairs, not shared-capacity
23 inventory control, and does not estimate implementation cost or planner
24 adoption.

25 The contribution of this paper is an evaluation protocol rather than a
26 new inventory model or solution algorithm. The comparator policies are
27 standard textbook constructions; what the paper adds is a reusable, au-
28 ditable procedure for deciding whether LLM-emitted policy values are ad-
29 missible and how they perform against ERP-derived baselines on a common
30 simulation horizon. Three components make up that protocol.

31 **C1 Admissibility protocol for LLM-emitted policy artifacts.**

32 Material-specific inputs pass through a configured enterprise LLM ap-
33 plication without project-specific training or fine-tuning. Deterministic
34 scoreability and capacity checks decide admission; the architecture
35 reserves a planner release step, which this study documents as a design
36 requirement rather than an evaluated intervention [12].

37 **C2 Executable boundary specification.** The paper documents the
38 material-input contract, alias-aware conversion, capacity validation,
39 and simulation boundary, so that a competing generator can be sub-
40 stituted and scored under identical rules.

41 **C3 Complete-cohort reporting discipline.** All 365 eligible pairs from
42 233,078 parsed records yielded scoreable, capacity-feasible artifacts af-
43 ter retry handling; 346 pairs meet the common feasibility rules used

44 for the three-arm descriptive comparison. No pair is silently dropped,
45 and every exclusion rule is prespecified and reported.

46 The remainder of the paper reviews related work, presents the executed
47 workflow and policy logic, reports exploratory post-cutoff simulation results,
48 and discusses the controls needed before a valid deployment evaluation.

49 **2. Literature Review**

50 Operations research has long addressed supply-chain uncertainty
51 through stochastic programming, robust optimization, and scenario-based
52 planning [4, 5, 6]. These methods formalize variability and recourse, but
53 operational inventory teams often face a more direct implementation prob-
54 lem: maintaining reorder points, safety stock, safety lead time, and order
55 quantities from incomplete historical evidence in ERP-driven planning en-
56 vironments [13, 14, 2, 3]. Static rules can be too coarse, while sophisticated
57 models can be difficult to explain and maintain. This motivates decision
58 support that keeps OR discipline while making material-level parameter
59 setting more transparent and auditable [7, 15].

60 Recent LLM research proposes natural-language interfaces, tool-using
61 agents, and pipelines that translate natural-language requests into optimiza-
62 tion model structures or executable code [9, 16, 17, 18]. Published concep-
63 tual systems and recent preprints illustrate a rapidly evolving frontier, in-
64 cluding multi-agent LLM systems that make autonomous per-period inven-
65 tory decisions [19, 20, 21]. They do not by themselves specify how inventory
66 histories should be reduced into demand regimes, lead-time diagnostics, ser-
67 vice buffers, and feasible policy parameters. Prior work therefore supports
68 the hybrid position adopted here: LLM artifacts can structure workflows

69 and explanations, while deterministic parsing and simulation make their
70 consequences inspectable and planners retain decision authority [9, 10].

71 InventOR follows this hybrid view. It supplies one filtered historical CSV
72 for the focal material plus an inline parameter block through an enterprise
73 LLM application. The intended prompt specification prohibits stochastic
74 simulation, machine-learning libraries, and unrestricted data fabrication,
75 but the stored run metadata do not independently identify the deployed
76 prompt or runtime. Because LLM outputs can contain fabricated or unfaith-
77 ful content [22], the executed downstream parser uses alias-aware lookups
78 and nested-field resolution before requiring numeric positive reorder point
79 and order quantity, numeric non-negative safety stock, and a resolvable sim-
80 ulator family. No project-specific model fitting or fine-tuning is performed.
81 The design treats the LLM as an artifact-generation component inside a
82 bounded decision process. Table 1 positions the framework relative to re-
83 cent AI-supported OR and inventory-control work.

Table 1: Literature positioning: InventOR relative to recent AI-supported OR contributions for inventory and supply chain decision support.

Study	LLM role	Policy class	Demand model		Analytical result		Evaluation
Huang et al. [16]	NL→code	General OR	Not fied	speci-	None		Benchmark
Ahmaditeshnizi et al. [18]	NL→model	General OR	Not fied	speci-	Solver-linked		Benchmark
Zhang et al. [20]	Reasoning agent	General OR	Not fied	speci-	Model solve	and	Case studies
Quan and Liu [21]	Multi-agent control	Inventory	Sequential demand		Per-period decisions		Simulation
Prak et al. [23]	None	Inventory control	Compound Poisson / intermittent		Robust estimation	esti-	Empirical study
Babai et al. [24]	None	Forecasting-linked inventory	Regime-specific		Review thesis	syn-	Review
Wasserkrug et al. [9]	Interface	General	Not fied	speci-	None		Conceptual
Cohen et al. [17]	Vision	General SCM	Not fied	speci-	None		Vision statement
InventOR	Prototype	(r, Q) executed; (s, S) implemented, unexercised	Artifact-defined values		Descriptive		Post-cutoff simulation

84 The table highlights the paper’s narrower contribution: material-level
85 parameter artifacts evaluated by a deterministic policy-family simulator.
86 Unlike InvAgent’s sequential multi-agent decisions, InventOR emits one sta-
87 tionary policy per material from a bounded historical package; unlike Opti-
88 MUS and OR-LLM-Agent, it does not formulate or solve general optimiza-
89 tion models.

90 3. Methodology

91 The unit of analysis is a plant-material pair. The framework records
92 a chain of custody from historical records to planner-facing policy values
93 by separating deterministic input preparation, configured LLM inference,
94 schema parsing, deterministic simulation, and human review [7, 25]. Each

LLM session receives one filtered, material-scoped historical CSV as a session resource and a small inline parameter block containing current cost, MOQ, service, SAP reference, and material-level storage-capacity fields. No project-specific model weights are trained, fine-tuned, or updated. Because the stored metadata omit prompt hash, application version, and model/runtime identifier, this study does not claim that the exact deployed instruction configuration is independently reproducible. Supplementary File S1 reproduces the intended application-side system and tool prompt texts verbatim, together with their disclosed discrepancies against the executable simulator.

3.1. Problem Definition: Single-Item Continuous-Review Control with Stress-Adjusted Service Buffering

The executed task is a single-item, single-echelon continuous-review simulation with lost sales [13, 14, 26]. Each plant-material pair is treated independently; multi-item coupling, shared capacity, substitution, and production scheduling are outside the pilot. For the SAP-SLT-informed OR comparator, pre-cutoff working-day demand yields mean demand $\bar{d}_i$, annualized demand $\bar{D}_i^{\text{ann}} = 250\bar{d}_i$, lead-time-demand dispersion, and stationary (R_i, Q_i) parameters. This comparator is not independent of ERP inputs because source SAP safety lead time enters effective lead time. For the SAP-derived arm, source safety stock is retained while R_i and Q_i are recomputed by the same executable routines. For the LLM arm, scoreable policy parameters originate directly in the artifact and pass alias-aware compatibility parsing plus strict executable-contract validation; the reference equations below do not reconstruct them. The simulator executes validated (r, Q) artifacts as fixed-quantity policies and validated (s, S) artifacts as order-up-to

121 policies. In this study all 365 scoreable artifacts resolved to the (r, Q) fam-
 122 ily after validation, so the order-up-to path is an available but unexercised
 123 capability rather than an evaluated result.

124 *Reference policy calculation.* The following equations describe the executed
 125 SAP-SLT-informed OR comparator; they do not reconstruct direct LLM-
 126 emitted policies. The order quantity $Q_i > 0$ is rounded upward to an integer,
 127 floored at source MOQ, and then projected to its material-level storage limit
 128 when possible. The reorder point $R_i \geq 0$ triggers replenishment. The EOQ
 129 anchor minimizes the standard annual cost approximation

$$\min_{Q_i} TC_i(Q_i; SS_i) = h_i \left(\frac{Q_i}{2} + SS_i \right) + K_i \frac{\bar{D}_i^{\text{ann}}}{Q_i},$$

130 where SS_i is arm-specific safety stock, $h_i = \text{holding_cost_rate}_i \cdot$
 131 unit_cost_i , and $K_i = \text{ordering_cost}_i$. The implementation first uses
 132 $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i(250\bar{d}_i)/h_i} \rceil\}$ when demand and ordering cost are
 133 positive, and MOQ otherwise. For every arm, a supplied capacity requires
 134 $\text{MOQ}_i \leq Q_i$ and $SS_i + Q_i \leq \text{max_storage_units}_i$; comparator quantities
 135 above the residual limit are reduced, while pairs with $SS_i + \text{MOQ}_i$ above
 136 the limit are excluded from the common cohort. The SAP-SLT-informed
 137 OR safety stock is $SS_i = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil$, with $R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i \rceil$. The
 138 SAP-derived arm substitutes source SAP safety stock into the same reorder-
 139 point expression. No lot-size multiple or stress coefficient is executed. The
 140 primary holding/order cost has a zero shortage penalty; positive lost-sales
 141 penalties are sensitivity analyses.

142 *Core constraints.* Inventory evolves under a lost-sales continuous-review rule
 143 on the working-day decision set $\mathcal{W}$. All arms share the first observed post-

144 cutoff on-hand inventory; the SAP-SLT-informed OR $R_i + Q_i$ is used only
 145 when that observation is unavailable. The initial on-order pipeline is empty.
 146 Duplicate same-material, same-date rows are aggregated by summing de-
 147 mand, receipts, corrections, and scrap and averaging positive lead times,
 148 but the simulator consumes only demand and excludes observed receipts,
 149 corrections, and scrap. The simulator rounds mean source lead time and
 150 schedules every modeled order by that number of subsequent working ob-
 151 servations:

$$I_{i,t} = \max(0, I_{i,t-1} + a_{i,t} - d_{i,t}), \quad t \in \mathcal{W},$$

152 where $a_{i,t}$ is the quantity whose working-day due observation is no later than
 153 t . Scheduled arrivals are added first, realized demand is served subject to
 154 lost sales, and the post-demand inventory position is computed as on-hand
 155 plus outstanding orders. For a validated (r, Q) artifact, a fixed quantity is
 156 ordered when

$$\delta_{i,t} = \mathbf{1}\{\text{IP}_{i,t}^+ \leq R_i\}, \quad \text{IP}_{i,t}^+ = I_{i,t} + \sum_{s: \text{due}(s) > t} o_{i,s},$$

157 with $o_{i,t} = Q_i \delta_{i,t}$. For a validated (s, S) artifact, the same threshold test
 158 uses s_i , and an order of $S_i - \text{IP}_{i,t}^+$ is placed when positive; no artifact in
 159 this study triggered that branch. Orders due after the observed horizon
 160 remain in transit rather than being forced onto the final observation. Pa-
 161 rameters remain stationary. Shared observed initialization and the empty
 162 initial pipeline are part of the executed comparison design.

163 The reorder point targets a nominal cycle service level $\alpha_i \in [0.5, 0.999]$

164 after numeric clamping. Under the normal approximation,

$$\mathbb{P}\left(D_{i,L_i^{\text{eff}}} \leq R_i\right) \geq \alpha_i \iff R_i \geq \bar{d}_i L_i^{\text{eff}} + z_{\alpha_i} \sigma_{LTD,i}.$$

165 The reported backtest metric is the ex-post fill rate, which is distinct from
 166 the nominal cycle-service target because it also depends on Q_i and realized
 167 demand [27].

168 Lead-time-demand variance is propagated under the demand and lead-
 169 time independence approximation,

$$\sigma_{LTD,i}^2 = L_i^{\text{eff}} \sigma_{D,i}^2 + \bar{d}_i^2 \sigma_{L,i}^2.$$

170 The executed SAP-SLT-informed OR buffer is

$$SS_i^{\text{OR}} = z_{\alpha_i} \sigma_{LTD,i},$$

171 with values rounded upward:

$$SS_i^{\text{OR}} = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil, \quad R_i = \left\lceil \bar{d}_i L_i^{\text{eff}} + SS_i^{\text{OR}} \right\rceil.$$

172 The executable statistics add non-negative source SAP safety-lead-time days
 173 to mean lead time when constructing comparator lead-time-demand disper-
 174 sion (Section 3.5 discusses this confound). LLM-emitted values remain ar-
 175 tifact values accepted by alias-aware parsing and strict executable-contract
 176 validation.

177 *Parser and review boundary.* Artifact policy labels determine simulator
 178 routing after validation. The parser resolves compatibility aliases, then
 179 requires a valid (r, Q) tuple or explicit s and S values for an (s, S) label;

180 it rejects malformed, infeasible, or internally inconsistent artifacts. It does
181 not execute label-specific optimization or deterministic escalation. Because
182 no artifact supplied explicit and internally consistent s and S values, every
183 scored policy in this study was executed as (r, Q) ; the reported evidence
184 therefore concerns fixed-quantity control only. Planner review remains
185 required before any operational use.

186 Methodologically, the contribution is therefore not to replace OR with an
187 LLM. Rather, it is an integration of deterministic input preparation, config-
188 ured LLM artifact generation, schema parsing, and explicit policy simulation
189 in a single prototype workflow [10, 11].

190 3.2. *Historical Input Schema and Canonical Analysis Table*

191 The source table is grouped by plant and material, but the uploaded
192 material CSV contains only the subset listed in Table 2. Plant and material
193 identifiers are carried by the run key and inline block rather than repeated in
194 the uploaded rows. The inline block also passes current cost, MOQ, service,
195 SAP reference, working-day demand, lead-time, and material-level storage-
196 capacity variables. Deterministic preprocessing supplies input normaliza-
197 tion, eligibility filters, and simulator statistics. Stock-flow fields omitted
198 from the uploaded subset are not analyzed by the executed LLM path.

Table 2: Fields included in the material-scoped CSV uploaded by the executed runner.

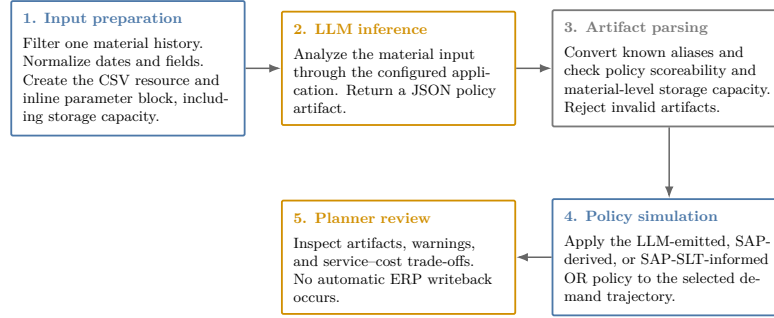
Column	Type	Description and method use
date	date / datetime	Calendar observation date; time ordering, aggregation, and diagnostics
actual_demand	numeric	Realized demand; mean, variability, intermittency, trend, and regime classification
delivery_time	numeric	Recorded replenishment lead time; rounded and scheduled as subsequent working observations by the simulator
minimum_order_quantity	numeric	Minimum replenishment quantity; order-size feasibility constraint
safety_stock_units	numeric	Existing safety stock; benchmarking against computed recommendations
safety_time_ workdays_actual	numeric	Source SAP safety-lead-time field; currently enters comparator effective lead time
supplier_plant_ distance	numeric	Supplier-to-plant distance; contextual lead-time-risk information
reliability_calc	numeric	Source supplier or lane reliability context
transportation_network	string	Source transportation-network context
service_level_target	numeric	Source service target used by comparator calculations
unit_cost	numeric	Purchase or standard cost per unit in the source dataset’s internal cost denomination; inventory-value proxy and input to holding-cost derivation and EOQ calculation
holding_cost_rate	numeric	Annual carrying-cost rate as a decimal (e.g., 0.20 for 20% of unit value); used to derive the annual cost of carrying one unit in stock, balance replenishment frequency against stock exposure, and compute economic order quantity
ordering_cost	numeric	Fixed replenishment cost in the source dataset’s internal cost denomination; input to EOQ calculation and simulated material-horizon holding/order cost
max_storage_units	numeric	Material-level upper bound applied to all evaluated arms: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
is_working_day	binary	Working-day flag; working-day demand statistics and non-zero demand share

3.3. Analytical Workflow

The workflow transforms a historical material snapshot into a policy artifact through deterministic preparation, LLM inference, parsing, and simulation (Figure 1).

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: Prototype workflow from material input preparation to policy artifact and simulation output.

203 The runner uploads a normalized material CSV and inline block; the
 204 configured application returns a JSON artifact. The parser converts ac-
 205 cepted fields to an executable (r, Q) or (s, S) policy and rejects malformed
 206 or infeasible outputs. Missing or invalid artifacts are recorded and retried.
 207 The simulator evaluates accepted LLM values, SAP-derived values, and the
 208 SAP-SLT-informed OR policy on the common horizon. An aggregate-only
 209 manifest records active-input, evaluation-code, and stored-artifact hashes;
 210 it explicitly records unavailable historical deployed prompt, application,
 211 model/runtime, setting, and timestamp metadata. Current runner code
 212 records these fields for future runs when the corresponding environment vari-
 213 ables are supplied. The author declares retrospectively that the evaluated
 214 artifacts were produced with an Anthropic Claude Sonnet 4.6 model behind
 215 the enterprise application, using a code-interpreter style Python execution
 216 tool; this declaration is recorded in the manifest as unverified because no
 217 deployed model identifier or generation settings were persisted at run time,
 218 and it must not be read as machine-verified provenance. Stored inputs,

artifacts, and code reproduce a recorded simulation, but not a fresh LLM completion; planners retain implementation authority.

3.4. *Artifact Completeness and Future Readiness*

The completeness audit asks whether each eligible pair has a scoreable artifact after retries; it does not estimate first-pass reliability, explanation completeness, or deployment readiness. The empirical section is a retrospective audit of an already-executed workflow rather than a controlled experiment: the generation cohort was produced before the evaluation protocol was finalized, the deployed configuration was not frozen or instrumented in advance, and no treatment was assigned. The post-cutoff simulation uses artifacts generated exclusively from pre-cutoff inputs and is descriptive rather than a controlled prospective trial [28, 29]. A future evaluation must freeze the deployed configuration, model/runtime identifiers, access controls, and approval workflow.

3.5. *Safety Lead Time Computation*

Safety lead time (SLT) is an ancillary ERP-facing design field. The prototype specification proposes combining transport, supplier, and quality lead-time variability with a reliability-deficit premium [13, 3], but the present release does not document estimators for those components. In the executed comparator code, non-negative source SAP safety-lead-time days are added to mean lead time before computing both SAP-derived and OR-computed dispersion and reorder points (this coupling is also flagged in Section 3.1). The OR arm is therefore labelled SAP-SLT-informed, and no separate SLT effect is identified.

243 3.6. *Post-Cutoff Policy Simulation*

244 The confidential operational extract contains 233,078 parsed CSV
245 records, 411 plant-material pairs, and three anonymized plants (Plant
246 A, Plant B, and Plant C). The parameter cutoff is 2026-03-10 and the
247 post-cutoff simulation horizon ends on 2026-08-27. Eligibility filters retain
248 365 plant-material pairs with sufficient pre-cutoff and post-cutoff evidence:
249 234 from Plant A, 65 from Plant B, and 66 from Plant C. Plant C has
250 no positive source working-day flags, so its calendar flags are derived
251 from same-date Plant A and Plant B votes; ties classify as working
252 days. This deterministic repair permits a common decision calendar;
253 because no alternative Plant C calendar exists, robustness is examined by
254 reporting calendar-day arrivals and by re-running the full comparison on
255 the native-calendar subset that excludes Plant C entirely. SAP-derived and
256 SAP-SLT-informed OR parameters use pre-cutoff information. Stored LLM
257 artifacts were generated from cutoff-bounded pre-cutoff inputs, so every
258 arm is evaluated on the same post-cutoff horizon. Nineteen pairs cannot
259 meet comparator safety stock plus MOQ within their supplied capacity;
260 these are excluded before the common 346-pair comparison.

261 The simulated arms are: (i) SAP-derived, retaining source SAP safety
262 stock while recomputing reorder point and EOQ order quantity from pre-
263 cutoff statistics; (ii) SAP-SLT-informed OR, using the same statistics, source
264 safety-lead-time input, and feasibility rules but without LLM inference; and
265 (iii) LLM-emitted, a validated (r, Q) or (s, S) artifact. The simulator is
266 continuous-review and lost-sales: all arms begin with the same first observed
267 post-cutoff on-hand inventory, using the OR $ROP + Q$ only as a shared fall-
268 back. The initial on-order pipeline is empty. On each working row, modeled
269 arrivals due after rounded mean working-day lead time are received, demand

is served subject to lost sales, post-demand inventory position is checked, and a policy-specific order is placed if its threshold is met. Observed post-cutoff receipts, corrections, and scrap are excluded. This common startup design may create cutoff transients. Metrics are fill rate, aggregate demand-weighted fill rate, materials above 95% fill, zero-stockout materials, stockout days, average on-hand inventory, and simulated material-horizon cost.

For policy p and material i , the cost summaries use the simulated horizon of T_i working days. Let $\bar{I}_{i,p}$ be average on-hand inventory, h_i annual holding cost per unit, K_i fixed ordering cost, and $n_{i,p}^{\text{orders}}$ simulated orders. The implementation uses a constant 250 annual working days and computes

$$TC_{i,p} = h_i \bar{I}_{i,p} \frac{T_i}{250} + K_i n_{i,p}^{\text{orders}}.$$

This is a simulated material-horizon holding/order cost in the source dataset’s internal cost denomination, not an audited annual financial or booked accounting result. Table 3 reports arithmetic means across the cohort with zero shortage penalty. Sensitivities add lost-sales penalties of one and five times source unit cost; cost values must be read jointly with fill rate and stockout days.

4. Results

4.1. Completion and Artifact Validity

The final execution produced parser-scoreable artifacts for all 365 eligible plant-material pairs. The simulation artifact reports zero missing and zero parser-rejected outputs after retry handling. The parser uses compatibility containers, alias lookups, and nested-field resolution before requiring numeric positive reorder point and order quantity, numeric non-negative

safety stock, a resolvable simulator family, and compliance with the supplied material-level storage bound. This establishes policy scoreability and capacity feasibility for the deterministic simulator, not full conformance to every field requested by the prompt. Eventual completeness after targeted retries is not an estimate of first-pass reliability or production availability.

4.2. Main Three-Arm Exploratory Simulation

Table 3 reports three policy arms on the same 346 common-feasible material trajectories. The SAP-derived arm retains source SAP safety stock but recomputes reorder point and EOQ quantity from pre-cutoff statistics; it is not the unchanged incumbent SAP policy. SAP-SLT-informed OR evaluates an (r, Q) comparator whose effective lead time retains source SAP safety lead time. The LLM-emitted arm evaluates (r, Q) or (s, S) artifact values generated from cutoff-bounded pre-cutoff inputs after alias-aware validation and material-level capacity checks. All arms satisfy the same MOQ and supplied per-material storage rule. Nineteen otherwise eligible pairs whose comparator safety stock plus MOQ exceed capacity are excluded. The comparison remains descriptive simulation rather than controlled production trial.

Table 3: Post-cutoff working-day simulation on 346 common-feasible plant-material pairs. All arms use pre-cutoff parameterization or cutoff-bounded inputs and satisfy identical MOQ and supplied storage rules. Cost is mean simulated material-horizon holding/order cost with zero shortage penalty. SO days is the total count of stockout days across the cohort.

Policy	n	Mean FR (%)	Agg. FR (%)	$\geq 95\%$ (%)	Zero SO (%)	Mean cost	SO days
SAP-derived	346	98.38	79.99	94.51	80.35	3,184.20	741
SAP-SLT	346	98.27	79.58	93.93	88.73	3,852.22	799
OR							
LLM-emitted	346	97.94	77.52	92.49	81.50	2,848.96	936

Relative to SAP-derived, LLM-emitted policies have 2.48 percentage points lower aggregate fill, 335.24 lower mean holding/order cost, and 195

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.

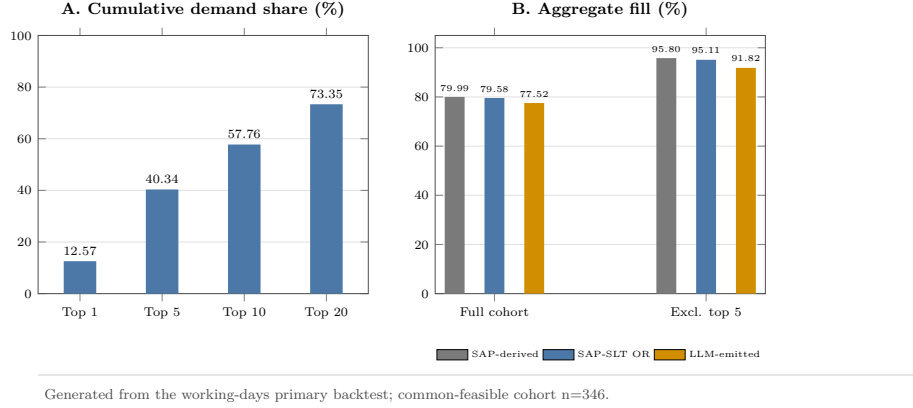


Figure 2: Demand concentration and aggregate-fill sensitivity. Panel A reports cumulative demand share for the largest materials. Panel B compares full-cohort aggregate fill with the value after excluding the five largest-demand materials.

312 additional stockout days. Relative to SAP-SLT-informed OR, they have 2.07
 313 percentage points lower aggregate fill, 1,003.26 lower mean holding/order
 314 cost, and 137 additional stockout days. These values describe one simulated
 315 cohort, not a definitive ranking or an economic preference.

316 Aggregate evidence is demand-concentrated. The largest 1, 5, 10, and
 317 20 materials contribute 12.57%, 40.34%, 57.76%, and 73.35% of post-cutoff
 318 demand. Excluding the five largest-demand materials raises aggregate fill
 319 to 91.82% for LLM-emitted, 95.80% for SAP-derived, and 95.11% for SAP-
 320 SLT-informed OR, versus 77.52–79.99% in the full cohort (Figure 2). Thus,
 321 the largest-demand items drive aggregate fill and should not be obscured by
 322 cohort averages.

323 At material level, LLM-emitted fill is not lower than SAP-derived for
 324 314 of 346 materials (90.8%), while its cost is no higher for 238 (68.8%);
 325 both conditions hold for 211 (61.0%). Against SAP-SLT-informed OR,
 326 the corresponding counts are 309 (89.3%), 303 (87.6%), and 269 (77.7%).

327 Paired LLM-minus-comparator fill differences have first quartile, median,
 328 and third quartile of 0.00 percentage points for both comparators, reflecting
 329 many exact fill ties. Corresponding cost differences are -259.24 , -47.17 ,
 330 and 34.96 versus SAP-derived and -787.83 , -209.32 , and -25.26 versus
 331 SAP-SLT-informed OR. A deterministic 10,000-resample paired bootstrap
 332 gives LLM-minus-SAP fill and cost means of -0.44 percentage points (95%
 333 interval $[-0.99, -0.02]$) and -335.24 (95% interval $[-629.16, -105.29]$);
 334 against SAP-SLT-informed OR, the corresponding values are -0.33
 335 percentage points ($[-0.89, 0.14]$) and $-1,003.26$ ($[-1,363.81, -705.03]$).
 336 These descriptive resampling intervals and aggregate results jointly show
 337 service-cost trade-offs rather than uniform policy ranking.

338 The artifact audit covers all 365 scoreable policies: normalized policy-
 339 family agreement is 90.68%, and 42 artifacts meet a prespecified diagnostic-
 340 outlier rule (CV absolute error above 0.75 or LLM-to-pipeline safety-stock
 341 ratio outside $[0.5, 2.0]$). Removing the 23 flagged artifacts present in the
 342 common cohort raises LLM aggregate fill from 77.52% to 79.55%, while
 343 SAP-derived reaches 80.08% and SAP-SLT-informed OR reaches 79.70%;
 344 this post-hoc diagnostic subset does not establish a causal correction. Re-
 345 placing primary working-day receipt offsets with calendar-day offsets gives
 346 aggregate fills of 77.77%, 80.25%, and 79.84%, respectively. Removing the
 347 one plant whose source working-day flags are entirely absent, and whose
 348 calendar is therefore imputed, leaves 280 native-calendar pairs with aggre-
 349 gate fills of 74.03% for LLM-emitted, 76.55% for SAP-derived, and 76.35%
 350 for SAP-SLT-informed OR, mean material-level fills of 98.19%, 98.71%, and
 351 98.54%, zero-stockout shares of 81.79%, 88.21%, and 90.71%, and mean
 352 holding/order costs of 2,893.68, 3,471.66, and 4,168.76. In this native-
 353 calendar subset both comparators retain higher fill and higher zero-stockout

354 shares than the LLM arm, so the imputed calendar does not manufacture the
355 observed service gap. With a lost-sales penalty of one times unit cost, mean
356 total costs are 1,688,922.75, 1,631,321.28, and 1,692,746.91, respectively; at
357 five times unit cost, they are 8,433,217.92, 8,143,869.56, and 8,448,325.64.
358 Thus, zero-penalty holding/order cost cannot establish economic superiority.
359 These sensitivities do not resolve empty-pipeline, excluded-flow, or calendar-
360 repair limitations.

361 5. Discussion

362 The current results document an applied AI/OR prototype on one
363 anonymized industrial dataset. The LLM artifacts are generated from
364 cutoff-bounded, pre-cutoff material histories, and all arms use the same
365 post-cutoff horizon. The SAP-SLT-informed OR comparator is not inde-
366 pendent of ERP inputs. The comparison remains descriptive rather than a
367 controlled prospective trial: the 365-material artifacts support auditing and
368 code-path verification, while the 346-pair common-feasible cohort supports
369 no definitive performance ranking.

370 One structural asymmetry limits how the observed service gap may be
371 interpreted. Both comparator arms are warm-started from ERP settings
372 that already embed accumulated planner judgment: the SAP-derived arm
373 retains the source SAP safety stock and recomputes only the reorder point
374 and order quantity, and the SAP-SLT-informed OR arm consumes the source
375 SAP safety-lead-time field. The LLM arm, by contrast, receives only a
376 material-scoped history and a parameter block and must construct every
377 policy value from that evidence alone. The observed aggregate-fill deficit
378 of roughly two percentage points is therefore a warm-start versus cold-start

comparison, not a like-for-like contest between a generator and an optimizer at equal information. An ERP-independent cold-start OR arm, computing safety stock from pre-cutoff demand and lead-time statistics without any source safety-stock or safety-lead-time input, is the comparator required to isolate generator quality; it is not available in the present release and is listed among the future-work requirements.

InventOR makes the input, artifact, and evaluation boundary inspectable without making LLM inference deterministic [10, 9]. An aggregate-only manifest hashes the active input, evaluation code, and stored artifacts, while explicitly identifying unavailable deployed prompt, application, model/runtime, setting, and timestamp records. A stored input, artifact, and simulator version reproduce a recorded result; a fresh completion may vary. The tracked historical 315-material snapshots are byte-identical rather than distinct generation replicates, so they cannot estimate generation variance. Missing-output lists, parser checks, retries, and material-level simulation rows keep incomplete execution visible [11, 30]. The workflow avoids project-specific training, but offers no evidence on setup effort, inference cost, or planner adoption.

The accepted policies satisfy a per-material storage bound, not a joint capacity constraint. They do not model shared capacity, substitution, or coordinated ordering. Their service–cost trade-off is not a replacement recommendation; a prospective trial with economically grounded shortage costs, repeated inference, and enforced planner review is required before adoption [13, 23].

Several limitations delimit the conclusions. All comparisons remain descriptive simulations; the paired bootstrap intervals characterize resampling variation across this cohort, not repeated LLM inference or a causal ef-

406 fect. Evidence comes from one anonymized dataset, one horizon, and one
 407 preserved full-cohort LLM generation; fresh completions may vary. An ear-
 408 lier unconstrained 315-material artifact set is retained for provenance, but
 409 its different cohort and validation end date preclude attributing differences
 410 to the storage rule. Demand is concentrated, with the five largest mate-
 411 rials representing 40.34% of demand. The calendar-day-arrival, lost-sales-
 412 penalty, and diagnostic-outlier sensitivities are limited checks, not replace-
 413 ments for a pipeline or flow-data reconstruction. The simulation begins with
 414 observed on-hand inventory but no outstanding-order pipeline and excludes
 415 observed receipts, corrections, and scrap. Plant C’s working-day calendar
 416 is reconstructed from other plants; the native-calendar subset excludes it,
 417 but no independently validated Plant C calendar is available. The SAP-
 418 SLT-informed OR arm retains an SAP input, and the LLM receives SAP
 419 reference context; neither comparator establishes independent incumbent-
 420 SAP performance. The capacity rule is per material and does not model
 421 shared capacity, substitution, multi-echelon interactions, or coordinated or-
 422 dering [31, 32]. Recommendation quality remains dependent on data quality,
 423 historical representativeness, application configuration, and policy conver-
 424 sion rules [15, 25]. Narrative completeness and planner effectiveness were
 425 not evaluated.

426 These limitations define the scope for future work: a prospective eval-
 427 uation that freezes the deployed configuration and model/runtime iden-
 428 tifiers, repeats inference, initializes the replenishment pipeline, validates
 429 lead-time and calendar semantics, includes unchanged incumbent and ERP-
 430 independent comparators, estimates economically grounded shortage costs,
 431 reports first-pass reliability, and extends to multi-item and shared-capacity
 432 settings with planner review. The present evidence supports a prototype

433 with traceable inputs, stored artifacts, and deterministic simulation; it does
434 not support a controlled-trial comparative performance claim [28, 29].

435 **6. Conclusion**

436 This paper presents *InventOR*, an applied AI/OR prototype that treats
437 LLMs as candidate-policy generators within deterministic preparation, pars-
438 ing, and simulation. The framework preserves an inspectable link between
439 material histories, policy fields, and simulated outcomes without project-
440 specific model training or fine-tuning.

441 The executed artifact set contains 365 scoreable policies generated from
442 cutoff-bounded, pre-cutoff material histories. Every accepted LLM policy
443 satisfies its material-level storage bound. On the 346-pair cohort where
444 all arms satisfy identical MOQ and supplied storage rules, LLM-emitted
445 aggregate fill is 77.52%, mean material-level fill is 97.94%, mean simulated
446 material-horizon holding/order cost is 2,848.96, and stockout days total 936
447 under working-day receipt scheduling. Positive shortage-cost sensitivities
448 remove the apparent cost advantage. These values are descriptive simulated-
449 comparison evidence, not repeated-inference or controlled prospective-trial
450 evidence. The framework establishes a reviewable prototype and protocol
451 for subsequent prospective evaluation, not a deployment or adoption claim.

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460 The author declares no known competing financial interests or personal
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462 paper.

463 **Data and Code Availability**

464 The raw operational data and material-level artifacts cannot be publicly
465 released because they contain commercially sensitive material, supplier, and
466 planning information. A confidentiality-screened release package contain-
467 ing analysis code, the anonymized variable schema, aggregate evaluation
468 results, manuscript sources, and aggregate integrity manifest is available
469 at github.com/DressPD/inventor_public. The release package is addi-
470 tionally deposited in a Zenodo archive whose version-specific DOI will be
471 registered at acceptance and cited in the final version of record; the GitHub
472 repository is the development mirror and the Zenodo deposit is the citable
473 archival copy. Supplementary File S1 accompanies this submission and con-
474 tains the intended application-side prompt specification. The historical de-
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485 Declaration of Generative AI and AI-Assisted Technologies in the 486 Writing Process

487 During manuscript preparation, the author used OpenAI GPT-5.6
488 through OpenCode for language editing, consistency checks, citation
489 review, and LaTeX proofing support; this is a writing-assistance tool and
490 is distinct from the Anthropic Claude Sonnet 4.6 deployment evaluated in
491 the research workflow. The author reviewed and edited the content after
492 use and takes full responsibility for the content of the published article.
493 Figures were rendered deterministically from author-reviewed LaTeX and
494 Python source; generative AI did not directly create or manipulate image
495 content. The research workflow evaluated in the paper is itself the object
496 of study and is described separately in the methodology.

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